Microeconomic Theory
Final Exam
2022

Instructions: Answering the questions in this exam is an individual effort. The estimated answering time is provided at the beginning of each question.

1. (Approximately 50 minutes)

   A consumer would like to buy a product from a monopolistic seller. The consumer's valuation of the good is v (thus v is her "type"), and the monopolist believes that v is uniformly distributed on $[0, 1]$. Assume that the consumer can send a message about her valuation to the monopolist, but she is restricted to either fully disclose her valuation (namely sending message v to the monopolist) or not to disclose at all (equivalently, she sends a message ND). After receiving the consumer's message, the monopolist chooses a price p to maximize her profits.

(a) Suppose just for this part that the consumer cannot send any message. What are the consumer's interim[1] payoff (or surplus) and the seller's optimal price and profits?

(b) Find a weak perfect Bayesian equilibrium (WPBE) in which all types of the consumer send message ND.

(c) Find a weak perfect Bayesian equilibrium (WPBE) in which all types of the consumer send message v.

2. (Approximately 60 minutes.) There are two states of the world $\{0, 1\}$. Both states are equally likely. States are not observable. In state 0, there are $N_0 = 2$ bidders; in state 1, there are $N_1 = 3$ bidders.

   Bidder i (for any i) observes her valuation v_i as private information. All valuations are uniformly and independently distributed in $[0, 100]$. Preferences and other details of the environment conform to the symmetric, independent private-values model.

(a) Show that the revenue equivalence theorm hold in this environment?

(b) Find a symmetric equilibrium strategy of the second-price, sealed-bid auction (SPA), and then find $U_i^{SPA}(v_i)$, the equilibrium interim utility of bidder i with type v_i.

(c) The seller runs a first-price, sealed-bid auction (FPA). Please find a symmetric equilibrium strategy of the FPA assuming that the equilibrim strategy is an increasing function. Do not proceed by solving a differential equation.

[1] That is, the expected payoff evaluated at the point that the consumer learn her type